405 Magnetic Fields

4.5.1 Interactions between magnetic poles

Like magnetic poles repel and unlike poles attract. Outside a magnet, field direction runs from north to south. Cutting a bar magnet produces smaller magnets, each with both poles, not isolated poles. Attraction of unmagnetised iron alone does not identify a pre-existing pole on the iron.

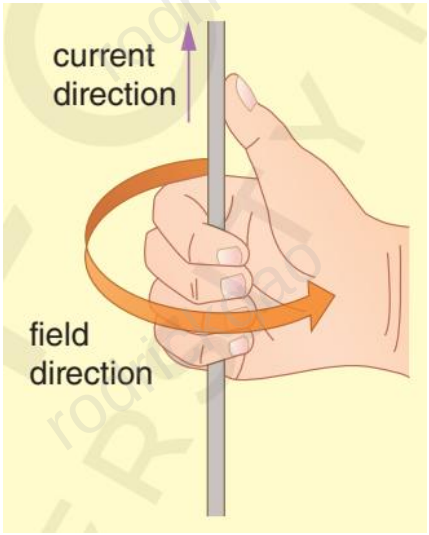
Textbook sources: 405 · PDF 2, 3, 4

4.5.2 · 4.5.3 · 4.5.8 Field around a long straight current

Field lines around a long straight current-carrying wire form concentric circles. Field strength increases with current and decreases with distance from the wire. Point the right thumb along conventional current; curled fingers show field direction. Electron flow is opposite. Reversing current reverses the field; use perpendicular distance from the wire in calculations.

$$B = \frac{\mu_0 I}{2\pi r}$$

Long straight wire in vacuum/air. B: T; I: A; perpendicular distance r: m; $\mu_0 = 4\pi \times 10^{-7}$ T m A^{-1}. Direction: right-hand grip rule.



Right thumb follows conventional current; curled fingers follow the circular magnetic field. Reverse the current and the field direction reverses.

Textbook sources: 405 · PDF 12, 20

4.5.4 Field of a circular coil

Different sections of a current-carrying circular coil reinforce the magnetic field at its centre, where the field is perpendicular to the coil's plane. Field lines form closed curves overall. Curl the right fingers along conventional current; the thumb gives the central field direction. Reversing current reverses this direction.

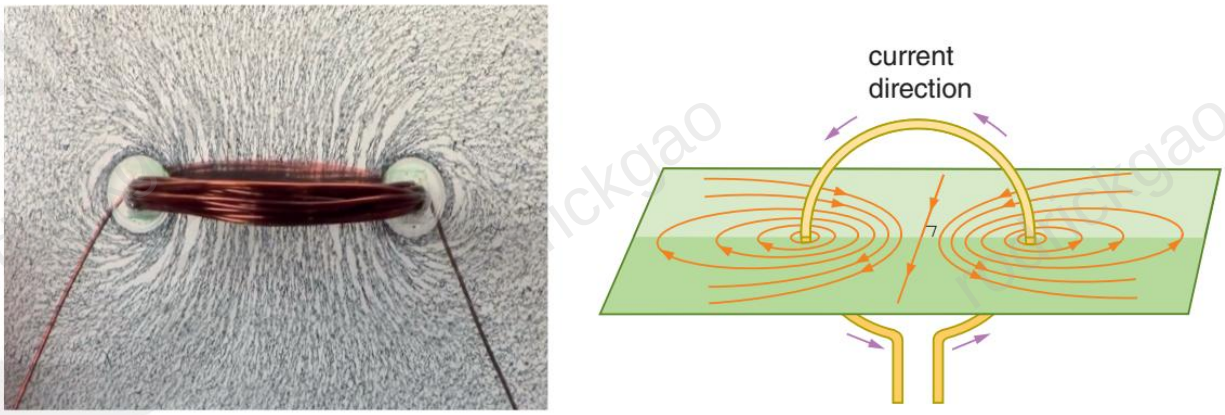


Fig 5.2e *Magnetic field pattern around a current-carrying circular coil.*

The coil field is perpendicular to the coil plane at its centre. Field lines form closed loops around the current.

Textbook sources: 405 · PDF 12

4.5.5 · 4.5.6 · 4.5.9 · 4.5.7 Solenoids and electromagnets

4.5.5 · 4.5.6 · 4.5.9 Field of a long solenoid

Inside a long solenoid away from its ends, approximately parallel, equally spaced field lines indicate a nearly uniform field. Its external pattern resembles a bar magnet. Curl right fingers along conventional current; the thumb points along the internal field towards the north pole. Strength depends on current and turns per unit length; the air-core formula does not directly apply to an iron core.

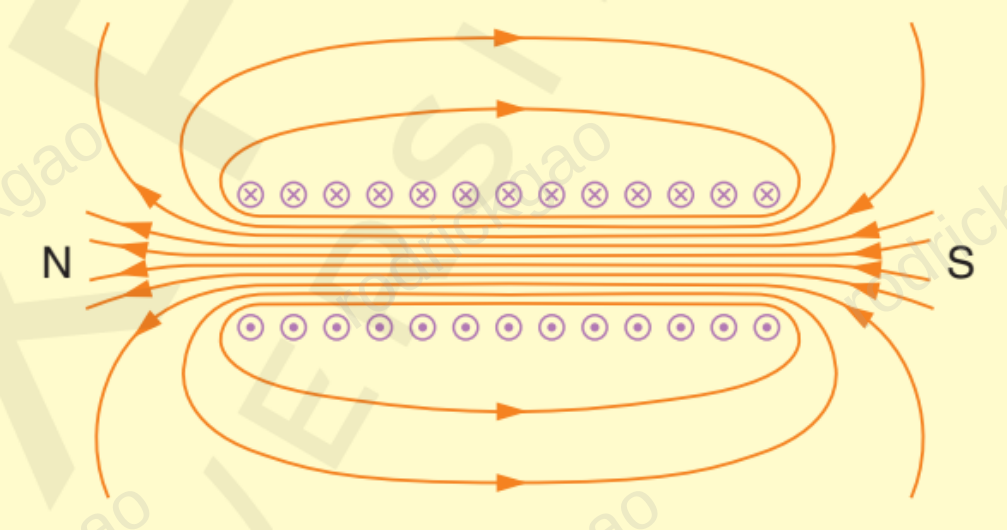
$$B = \frac{\mu_0 NI}{l} = \mu_0 nI$$

$$n = \frac{N}{l}$$

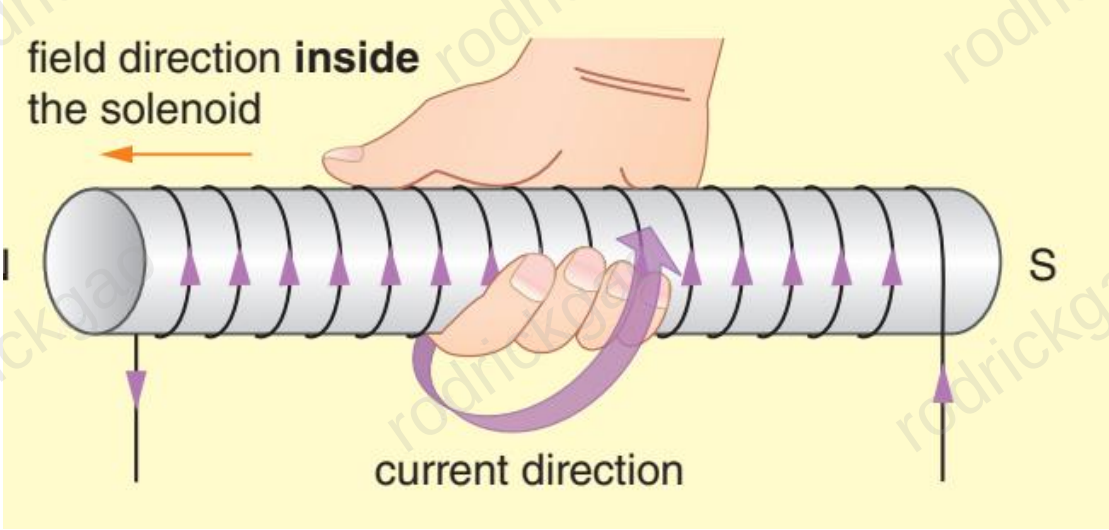
Inside a long air-core solenoid, away from ends. B: T; N: turns; l: length (m); n: turns per metre; I: A; $\mu_0 = 4\pi \times 10^{-7}$ T m A^{-1}.

4.5.7 Strengthening an electromagnet

For a solenoid of comparable geometry, increase field strength by increasing current, increasing turns per unit length or inserting a soft-iron core. Turn density matters, not total turns alone: lengthening the coil without adding turns may weaken the field. Soft iron magnetises and demagnetises readily, making it suitable for switchable electromagnets.



Inside the long solenoid the nearly parallel, evenly spaced field lines indicate an approximately uniform field. Outside, they run from N towards S.



Curl the fingers along conventional current around the turns. The thumb points along the internal magnetic field and towards the north end.

Textbook sources: 405 · PDF 13, 17, 24